Financial Conditions and the Stance of Policy

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Abstract

How much do financial conditions add to the stance of monetary policy beyond the policy rate? I measure conditions in policy units, as the move in the two-year yield that would bring the same change, and relative to the neutral rate r^*. The weights on the 10-year yield, equity valuation and the dollar come from their responses to monetary policy news in announcement windows. Over 1976–2026 the index explains unemployment a year ahead beyond the real funds rate relative to r^*, and the funds-rate gap adds nothing to the index. The information is in the 12-month change of the index, not in its level, and it is absent for core inflation. In 2004–06 the index read 1.29 points looser than the funds-rate gap, and in 2022–24 1.40 points. The index complements the Chicago Fed's NFCI, which carries the longer horizons and recession risk. Removing what lagged unemployment and inflation explain cuts its unemployment response by 45 to 67 percent. Put on a common basis (nominal, real, relative to r^*), the main indexes agree on direction and differ by about one point in level. The results hold on 1996–2026, with smaller effects.

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1 Introduction

Is monetary policy tight or loose? The policy rate does not answer by itself. Policy works through long yields, stock prices and the dollar, and these respond to much else, so the same funds rate can be more or less restrictive at different times. An index of financial conditions has long been proposed as a guide to the stance of policy, a counterpart of the Taylor rule (Hatzius et al., 2010), and policymakers describe policy this way. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023). The indexes do not always agree. In March 2024 the Chair said that "there are many different financial conditions indicators, and you can kind of, you know, see different answers to that question" (Powell, 2024).

I ask how much financial conditions add to the stance of policy beyond the policy rate, and whether the difference says anything about the economy ahead.

They add, and the addition is informative about unemployment. I measure financial conditions with a market index in policy units: a one-point move in the index is the change in conditions that a policy move raising the two-year yield by one percentage point brings. The index and the funds rate are both measured relative to the neutral rate r^*. Over 1976–2026, adding the index to the real funds rate relative to r^* raises the adjusted R^2 of the change in the unemployment gap a year ahead by 0.11 ($t = 5.49$), while adding the funds-rate gap to the index adds 0.00. The information is in the index's 12-month change, as it is in the Board's FCI-G, and not in its level. The index adds nothing for core inflation. In 2004–06 the index read 1.29 points looser than the funds-rate gap, and in 2022–24 1.40 points looser (Figure 1).

The index complements the Chicago Fed's NFCI instead of replacing it: the NFCI carries the longer horizons and recession risk, and the market index carries the next six to twelve months. Between 45 and 67 percent of the index's unemployment response is activity that lagged unemployment and inflation explain. The results hold on 1996–2026, the sample of the weights, at smaller size, and fail for payrolls. Put on a common basis, the major indexes agree on direction and differ by about one point in level.

The approach has three features. First, I build the index and put it on a common basis. Its weights are the responses of the 10-year yield, equity valuation and the real dollar to monetary policy news in narrow windows around FOMC announcements, combined to recover the news as precisely as possible. A two-variable VAR with the unemployment gap gives it a level and a flow. I measure every index in three steps, nominal rates, real rates and gaps to r^*, and at each step I compare like with like: my nominal index with nominal versions of the Goldman Sachs index, FCI-G and an index weighted with FRB/US, my real index with real versions, and my r^* index with r^* versions. Second, I compare the index with the real funds rate relative to r^* and ask whether it adds for unemployment, payrolls and inflation, in sample and out of sample, with the uncertainty of the weights and of r^* in the statements about episodes. Finally, I ask whether the flow is only activity reflected back, how it relates to the NFCI, whether the neutral-rate step and the weights matter for outcomes, and whether the results hold on 1996–2026 and with data as known at the time.

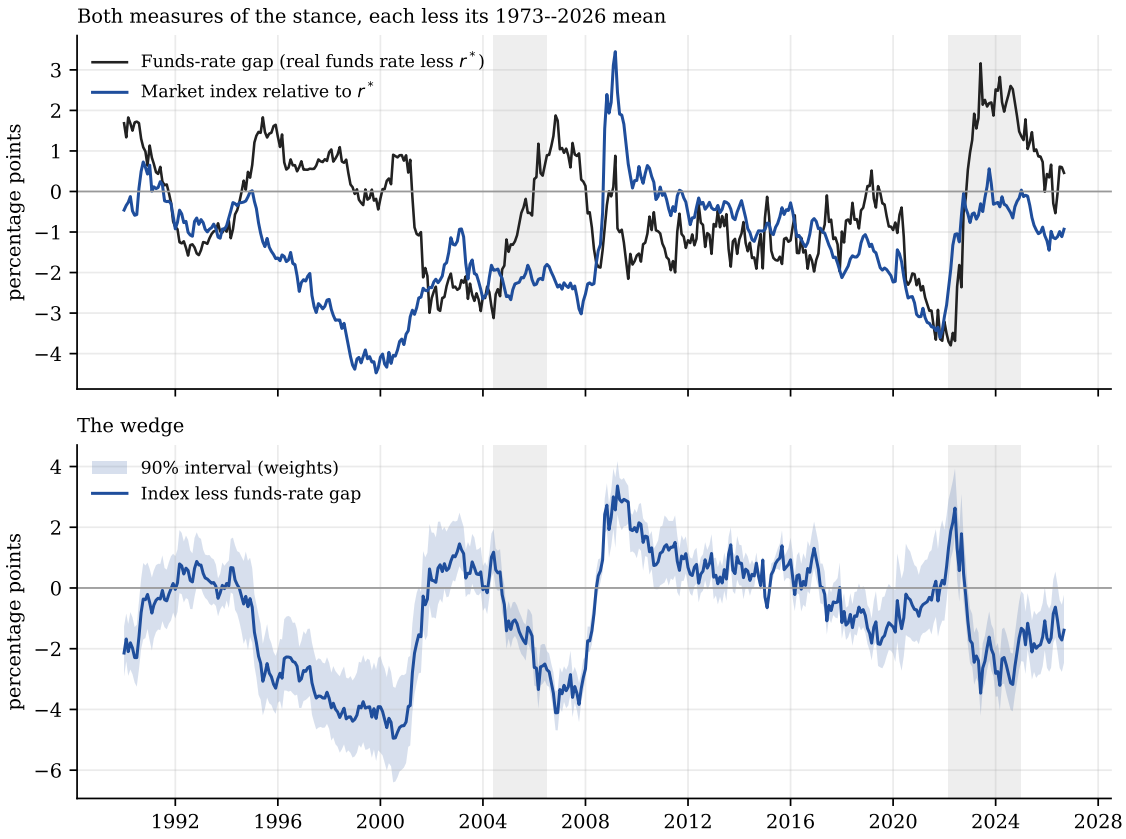


Figure 1: The Market Index and the Funds-Rate Gap

Notes: Monthly. The funds-rate gap is the effective funds rate less one-year expected inflation less r^* (Laubach–Williams); the market index is relative to r^* in two-year-yield points; each is shown less its 1973–2026 mean. The wedge is the index less the gap, with a 90 percent interval from bootstrap samples of the announcements that set the index's weights. Shaded: 2004–06 and 2022–24.

1.1 Connection to the Literature

The stance of policy. The first purpose of a financial conditions index in Hatzius et al. (2010) is to serve as one guide to the effective stance of policy after taking into account all other factors, as a Taylor rule does; their examples are the conundrum of 2004–06, when conditions stayed easy as the funds rate rose, and late 2008, when conditions were restrictive at a zero funds rate. Dudley (2010) adds that higher stock prices and faster productivity growth raise the equilibrium rate, so that the same funds rate is less restrictive. I measure the wedge between an index and the funds-rate gap in common units and test whether it matters.

Units and flows. Ajello et al. (2023) express conditions as their drag on GDP growth over the next year, with weights from FRB/US multipliers applied to changes, so that FCI-G is a flow with zero neutral for growth. Caballero, Caravello and Simsek (2026) define FCI*, the level of FCI-G consistent with output at potential, as the analogue of r^*. I express indexes in policy units, give the market index a flow of the FCI-G form, and measure rates relative to r^*; I do not estimate a neutral level for the index itself.

Stress, risk and the tail. The Chicago Fed's NFCI and ANFCI summarize 105 indicators of risk, credit and leverage by a dynamic factor model (Brave and Butters, 2011); Adrian, Boyarchenko and Giannone (2019) show that conditions shift the lower quantiles of growth. I treat them as a different kind of index, set the market index beside the NFCI, and report the lower tail in sample only.

Weights and comparisons. Weights come from effects on output (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018; Ajello et al., 2023), from principal components (Brave and Butters, 2011) or, as here, from event-study regressions (Kuttner, 2001; Gürkaynak, Sack and Swanson, 2005; Bernanke and Kuttner, 2005; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023). Aramonte, Rosen and Schindler (2013) compare twelve indexes by correlations of standardized series, Arnaut and Bauer (2024) estimate the response of the Goldman Sachs index to policy surprises, and Sack and Swanson (2023) relate FCI-G to policy-induced changes in the two-year yield. I put the indexes on one basis and ask what the market index adds to the stance, and I report the sampling error of its weights.

The neutral rate. What matters is the rate relative to the neutral rate, which fell from the 1990s to the 2010s (Laubach and Williams, 2003; Holston, Laubach and Williams, 2017, 2023; Hamilton et al., 2016; Bauer and Rudebusch, 2020; Kiley, 2020). I test whether measuring the inputs relative to r^* improves what the index says about the economy and report the answer as it is.

2 The Indexes Compared

Section 2.1 describes the existing indexes. Section 2.2 builds the market index, and Section 2.3 the index I weight with FRB/US. Appendix A gives the construction of every series.

Table 1: The Indexes

Index	Variables	Weights from	Unit	Reading
Market index	10-year yield, equity valuation (CAPE), real broad dollar	Responses to monetary policy news in announcement windows	Points of the two-year yield per unit of policy news	Level, and a flow from a VAR
GS (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018)	Federal funds rate, 10-year yield, BBB spread, equity valuation, nominal broad dollar	Published weights 4.4, 45.1, 39.6, 4.9 and 6.0 percent, from FRB/US and Goldman Sachs modeling	Index points	Level; zero is a base period
FCI-G (Ajello et al., 2023)	Federal funds rate, 10-year yield, mortgage rate, BBB yield, stock prices, house prices, dollar	FRB/US and other Board models	Percentage points of GDP growth over the next year	Flow, built from past changes; zero is neutral for growth
FRB/US index (mine)	Policy rate, 10-year yield, mortgage and BBB spreads, equity valuation, dollar; with and without house prices	FRB/US effect on GDP after four quarters	Funds-rate points	Level
NFCI, ANFCI (Brave and Butters, 2011)	105 indicators, mostly spreads, volatilities and credit quantities	Principal components (ANFCI: purged of activity and inflation)	Standard deviations	Level; zero is the sample average

Notes: GS is a replication with the published weights; the series itself is proprietary. The FRB/US index is my own construction. The market index is described in Section 2.2.

2.1 The Existing Indexes

Table 1 sets the indexes side by side. They differ in three ways that matter for a comparison.

The first difference is the source of the weights. The Goldman Sachs index and FCI-G take them from models of how each variable affects activity, the NFCI from the data's principal components, and my market index from how each variable responds to policy. The second is the reading. A level index says where conditions stand relative to a base period, as the Goldman Sachs index and the NFCI do. A growth index says what conditions imply for activity: FCI-G measures the contribution of financial conditions to GDP growth over the next year, and zero means that the effect of current and past changes adds nothing to growth and takes nothing from it. Zero in FCI-G is therefore neutral for growth, which is a different idea from a neutral level of conditions. FCI-G is built from changes and has no level. The third is the treatment of rates. The Goldman Sachs index uses nominal rates and the nominal dollar, so it trends over long spans (Hatzius et al., 2010). FCI-G uses nominal rates as well.

Units differ too. I convert the Goldman Sachs index and the NFCI and ANFCI to the market index's units by their own responses to the same policy news: the Goldman Sachs index rises by 1.04 points per unit of policy news, the NFCI by 0.097 standard deviations and the ANFCI by 0.19. FCI-G stays in percentage points of GDP growth and is compared through correlations and through its own scale.

Table 2: Market Index: Responses to Policy News and Weights

	Response	Weight	90% interval	Share of policy move
10-year Treasury yield (pp)	0.53 (0.06)	0.609	[0.128, 1.126]	0.33
Equity valuation, 100 ln CAPE (percent)	−7.78 (0.93)	−0.051	[−0.070, −0.029]	0.40
Real broad dollar, 100 ln (percent)	4.22 (0.53)	0.065	[0.021, 0.114]	0.28

Notes: Announcement windows, 253 announcements, January 1996 to September 2026. Response: change in the window per unit of policy news, which moves the two-year yield by 1 percentage point (heteroskedasticity-robust standard errors in parentheses). Weight: one-factor generalized least squares, normalized so that the weights applied to the responses sum to 1. Interval: 5th and 95th percentiles of weights re-estimated on 500 bootstrap samples of the announcements. Share: weight times response.

2.2 The Market Index: Weights from Policy News

Policy news. At each FOMC announcement I measure policy news with the factor of Nakamura and Steinsson (2018). It is the first principal component of the moves in five short-term interest-rate futures in a narrow window around the announcement, scaled so that one unit moves the two-year Treasury yield by one percentage point. The data are the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025), "monetary events" windows: 253 announcements from January 1996 to September 2026.

Responses and weights. In each window the 10-year yield, the equity valuation and the real dollar move by a response times the policy news plus a part that has nothing to do with it. The windows reveal the response ψ_j of each variable j. I choose the weights that recover the policy news most precisely from the three variables together: $w \propto \Sigma^{-1}\psi$, where Σ is the covariance of the parts that are not policy news, normalized so that $w'\psi = 1$ (one-factor generalized least squares). A policy move that raises the two-year yield by one percentage point therefore raises the index by exactly one unit, and the index is measured in points of the two-year yield.

Table 2 gives the results. A policy move that raises the two-year yield one point raises the 10-year yield by 0.53 points, lowers equity valuation by 7.78 percent and raises the real dollar by 4.22 percent. Each variable carries a substantial part of the policy move: the shares are 0.33 for the 10-year yield, 0.40 for equity valuation and 0.28 for the dollar. Applied to the demeaned levels of the 10-year yield, the equity valuation multiple (100 times the log of the cyclically adjusted price-earnings ratio, CAPE) and the real broad dollar, the weights give the index.

What the weights assume. Consider a rise in stock prices. If it comes from news about earnings, it has little to do with policy, and the index treats it as if it did, because it enters with the same weight as a rise that policy caused. If every movement in the three variables came from policy, the index would equal the cumulated policy-induced change in the two-year yield. The weights say what the variables do when policy moves, not what causes their movements at other times. An announcement can also reveal the Fed's view of the economy, and then the responses mix policy and information (Nakamura and Steinsson, 2018). I measure the responses to the news as the markets receive it, and I set the information effect aside. The index is a measure of the policy-equivalent

Table 3: FRB/US and Market Weights on the Three Common Variables

	FRB/US weight	Market weight	90% interval	FRB/US response	Market response
10-year yield (pp)	0.569	0.609	[0.128, 1.126]	0.37	0.53
Equity valuation (percent)	−0.032	−0.051	[−0.070, −0.029]	−13.18	−7.78
Real dollar (percent)	0.100	0.065	[0.021, 0.114]	1.62	4.22

Notes: FRB/US weight: effect on GDP after four quarters of a persistent unit shift in the variable, divided by the effect of a persistent 1 percentage point rise in the federal funds rate (0.4525 percent of GDP); funds-rate points per unit. Market weight: two-year-yield points per unit (Table 2); the interval is its bootstrap interval. FRB/US response: change in the variable four quarters after a persistent 1 percentage point funds-rate rise (VAR expectations). Market response: change in the announcement window per unit of policy news.

tightening in financial conditions. It is not a forecast of activity, and Section 4 gives it dynamics separately.

2.3 The Index Weighted with FRB/US

The main alternative to market weights is to weight each variable by its effect on activity in a macroeconomic model, as FCI-G does. For a comparison of levels I build such an index from FRB/US (Brayton, Laubach and Reifschneider, 2014). Each variable's weight is its effect on GDP after four quarters of a persistent unit shift, divided by the effect of a persistent one percentage point rise in the federal funds rate (0.4525 percent of GDP), so the index is in funds-rate points. It covers the policy rate, the 10-year yield, the mortgage and BBB spreads, equity valuation and the dollar, and in one version house prices. I also give it a flow in the form of FCI-G.

Table 3 compares the two sets of weights on the three common variables. They agree on the 10-year yield, where the FRB/US weight is 0.94 times the market weight. The model puts less weight on equity valuation (0.62 times) and more on the dollar (1.54 times). The two indexes are close on a pure policy move. The FRB/US weights price the response of the three variables to a unit of policy news at 0.97 funds-rate points, against exactly one point for the market index. They differ on movements that policy did not cause. The model gives a fall in equity prices its estimated causal effect on output, and the market index treats it as policy-equivalent. Because the two weight vectors are not proportional, the gap between the indexes is a weighted sum of non-policy movements in equity valuation and the dollar.

3 A Common Basis

The comparison runs in three steps, and each step answers a problem of the one before.

3.1 Nominal Rates

The first step uses rates as published. Every index is then a function of nominal yields, and the Goldman Sachs index and FCI-G also use the nominal broad dollar. A nominal index treats a

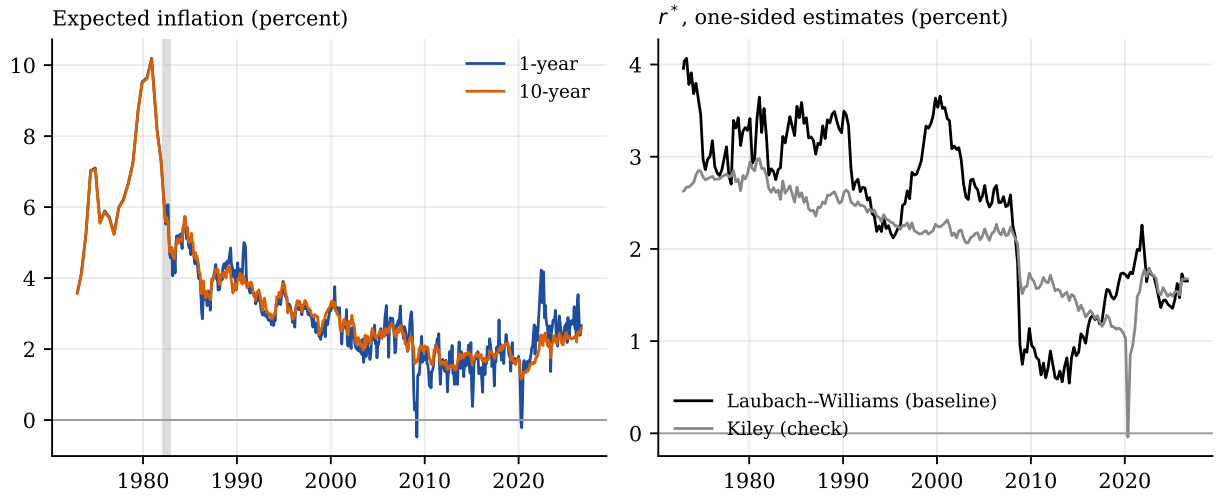


Figure 2: Expected Inflation and r^*

Notes: Left: expected inflation, Livingston Survey before 1982 and Cleveland Fed model estimates after, shifted to the Cleveland level over 1982–84 and blended over 1982; shaded: the blending year. Right: one-sided, full-sample estimates of r^* from the Laubach–Williams model (baseline) and Kiley's model (check). Percent.

10-year yield of 8 percent alike in 1981 and in 2000. That is reasonable only if inflation expectations and the neutral real rate are stable, and neither was. Expected inflation over the next year averaged 8.43 percent in 1981 and is 2.66 percent today, and the Laubach–Williams estimate of r^* averaged 2.76 percent over 1990–2007 and 0.86 percent over 2009–2016 (Figure 2). The nominal dollar poses a further problem for the Goldman Sachs index. Its broad index rises from 31 in 1973 to 119 today, mostly from inflation differentials, and this drags the Goldman Sachs index away from the others before 1990 (Figure 6, top panel).

3.2 Real Rates

The second step measures each rate less expected inflation at a matched horizon: the policy rate less one-year expected inflation, and longer rates less ten-year expected inflation. The dollar is the real broad index in every index, and in FCI-G stock and house prices are deflated by the CPI. This step removes the inflation trend. The nominal and real market indexes are almost the same over 1990–2026, with a correlation of 0.98, because inflation expectations were stable then. The real step matters before 1990. Nominal FCI-G shows a tightening in mid-1974 (0.60) that the real version does not (about zero), because nominal rates rose with expected inflation.

3.3 Gaps to the Neutral Rate

The third step measures each real rate relative to r^*: the policy rate, the 10-year yield, and in FCI-G the BBB and mortgage yields. Equity valuation enters through the earnings yield $E/P = 100/\text{CAPE}$,

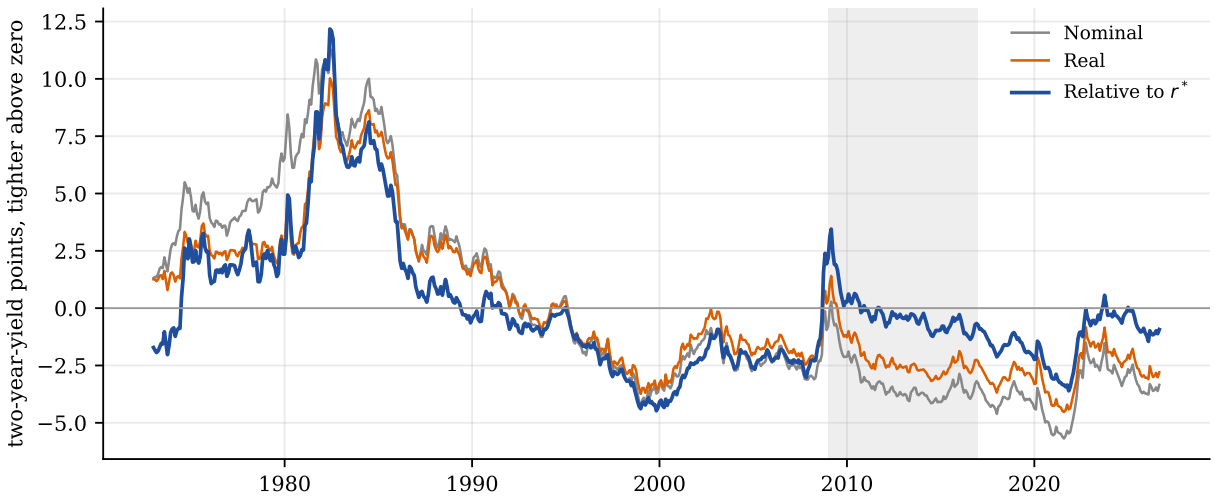


Figure 3: The Market Index at Each Step

Notes: Monthly, relative to the 1973–2026 mean, in two-year-yield points; tighter above zero. Shaded: 2009–2016.

which is also an implicit real rate, as the gap $E/P - r^*$. I convert each index's equity weight at the sample-average earnings yield, so that every index keeps its weight on CAPE. Spreads over the 10-year yield are unchanged, because r^* cancels. The baseline uses the one-sided, full-sample Laubach–Williams estimate, and Kiley's estimate is a check. Each variable is measured relative to its 1973–2026 average, so r^* matters through its movements over time.

3.4 What Each Step Changes

Figure 3 shows the market index at each step. The real step changes little after 1990, and the r^* step changes a lot: the correlation of the real and r^* versions over 1990–2026 is only 0.57. With r^* low after 2008, the step to r^* raises the 2009–2016 average of each index: by 1.87 points for the market index, 1.28 for the Goldman Sachs index and 1.52 for the FRB/US index. Relative to the 1990–2026 average, the market index averages 1.16 at the r^* step against -0.30 in the real version and -0.70 in the nominal one, so conditions in 2009–2016 look tighter than average rather than easier. The four level indexes (the market index, the Goldman Sachs index and the FRB/US index with and without house prices) are closer at each step: their average range over 1990–2026 falls from 1.15 points (nominal) to 0.97 (real) and 0.88 (relative to r^*).

4 Level and Flow

FCI-G measures a flow, and the other indexes measure a level. I give the market index both. The level is the index of Section 2.2. The flow comes from the relationship between the index and unemployment.

Table 4: Unemployment Response to a Tightening of the Market Index

	Lags	Granger F	p	$\Phi(6)$	$\Phi(12)$ [90%]	$\Phi(24)$	Months significant
Nominal	12	3.74	<0.0001	0.29	0.57 [0.29, 0.77]	0.66	4–36
Real	12	3.67	<0.0001	0.34	0.61 [0.32, 0.82]	0.68	3–36
Relative to r^* (baseline)	12	5.30	<0.0001	0.40	0.65 [0.39, 0.83]	0.64	1–36
Relative to r^* (Kiley)	12	4.63	<0.0001	0.37	0.61 [0.33, 0.80]	0.63	3–36
Relative to r^*, 4 lags	4	9.98	<0.0001	0.32	0.40 [0.21, 0.58]	0.38	1–36

Notes: Monthly VAR in the unemployment gap and the market index, January 1973 to September 2026, 2020 months with dummies; unemployment first. $\Phi(h)$: response of the unemployment gap (percentage points) h months after an innovation that raises the index by one unit. Granger: test that the lags of the index do not enter the unemployment equation. Interval: 5th and 95th percentiles of 500 block-bootstrap draws (12-month blocks of residuals). Months significant: first and last month with the 90 percent band above zero.

4.1 The Response of Unemployment

I estimate a monthly VAR in the unemployment gap (the unemployment rate less the Congressional Budget Office's natural rate) and the market index, from January 1973 to September 2026, with 12 lags and monthly dummies for 2020. Unemployment is ordered first, so it responds to the index only with a lag. I repeat the estimation for each step of the construction. Table 4 and Figure 4 give the results.

The index helps predict unemployment at every step (in the r^* baseline the Granger statistic is 5.30, p <0.0001). A one-unit tightening raises the unemployment gap gradually. In the baseline the response reaches 0.65 percentage point after 12 months, with a 90 percent interval from 0.39 to 0.83. With an Okun coefficient of two this is about 1.3 percent of output. I call the 12-month response the yardstick of the index, $\Phi(12)$. With four lags the response is smaller (0.40) and levels off earlier. The baseline has 12 lags; Section 8 reports four.

4.2 The Forward Impetus

With the response profile $\Phi(m)$ I compute what the changes in the index over the past year imply for the next 12 months, in the way FCI-G does with FRB/US multipliers:

$$\mathrm{FI}(t) = \sum_{m=0}^{11} \frac{\Phi(m+12) - \Phi(m)}{\Phi(12)} \, \Delta M(t-m).$$

Here M is the market index. A value of 1 means that the past year's changes raise unemployment over the next year by as much as a permanent one-unit tightening does in its first year. I call this the forward impetus. Figure 5 shows the level and the flow in the baseline. The flow correlates 0.84 with the r^* version of FCI-G, and the level only 0.30, because the two answer different questions.

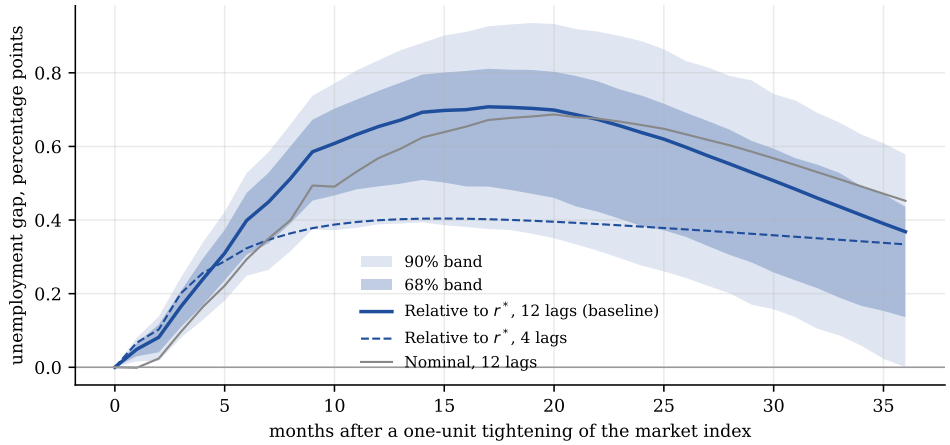


Figure 4: Unemployment Response to a One-Unit Tightening of the Market Index

Notes: Response of the unemployment gap after a shock that raises the index by one unit, in the monthly VAR with unemployment first (January 1973 to September 2026, 2020 months with dummies). Bands: 68 and 90 percent, from a block bootstrap (12-month blocks of residuals, recursive simulation).

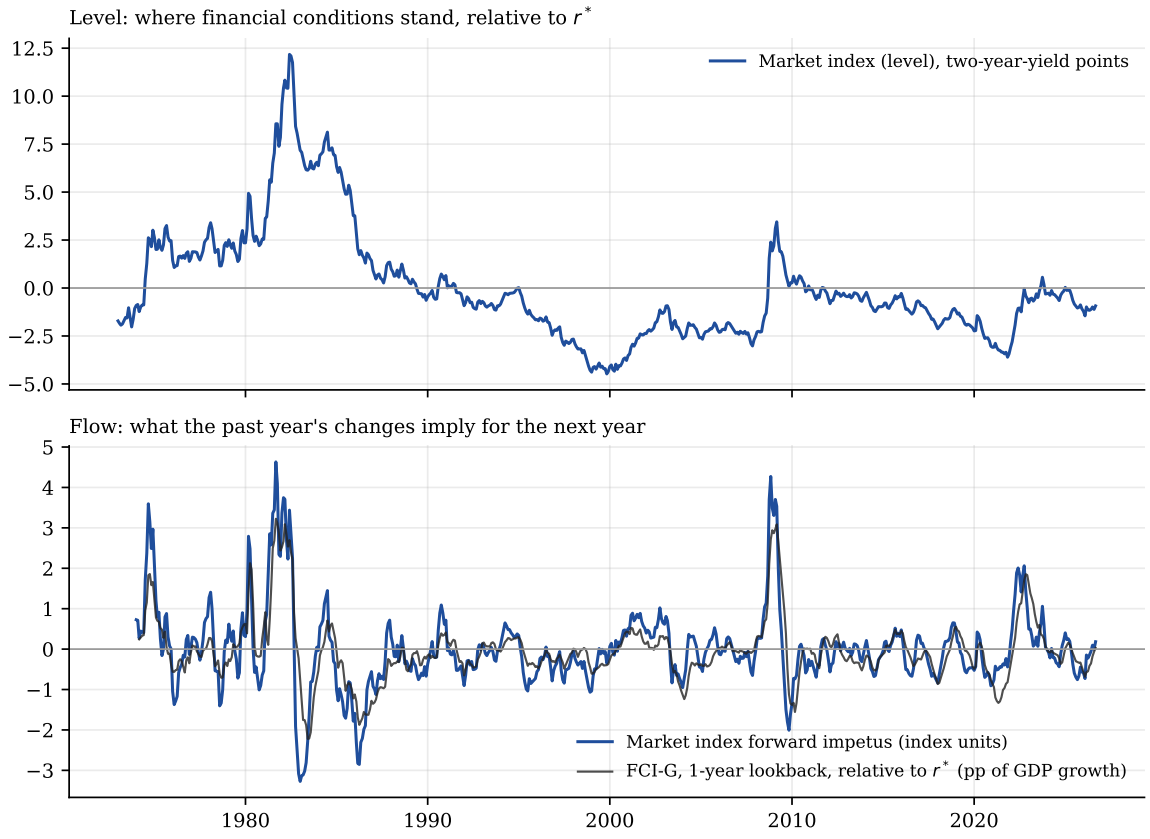


Figure 5: The Market Index: Level and Flow

Notes: Monthly, relative to r^*. Top: level in two-year-yield points, tighter above zero. Bottom: forward impetus of the 12-lag VAR (index units) and FCI-G with a one-year lookback in its r^* version (percentage points of GDP growth), on the same scale.

5 The Stance beyond the Policy Rate

5.1 The Funds-Rate Gap and the Wedge

The stance of policy read from the policy rate is the funds-rate gap: the effective federal funds rate less one-year expected inflation less r^*. The stance read from financial conditions is the market index relative to r^*. Each is measured relative to its 1973–2026 mean. They differ in units, two-year-yield points for the index and funds-rate points for the gap, and the comparison treats a point of each as a point of policy. The wedge is the index less the gap. A policy move that raises the two-year yield raises both. A rise in the dollar or a fall in equity valuation that policy did not cause raises only the index, and the wedge is the part of conditions that the funds rate does not carry. Levels of the two correlate 0.54 over 1976–2026 and their 12-month changes 0.31; they are -0.007 and -0.033 in 2004–06, when the funds rate rose and conditions did not follow.

5.2 Does the Wedge Matter for the Economy Ahead?

Table 5 adds the index to the funds-rate gap in projections of the change in the unemployment gap, payroll growth and core inflation 12 months ahead.

Unemployment. The funds-rate gap alone explains little (the t-statistic of its 12-month change is 1.46), and the index alone explains much (5.66). Given the gap, the index raises the adjusted R^2 by 0.11 in the flow form and by 0.12 when the gap enters as a level; given the index, the gap adds 0.00 and 0.02. In levels the index adds 0.00 ($t = 0.43$). The result holds for the Goldman Sachs index (gain 0.12) and the FRB/US index (0.10), and the t-statistic of the market index rises with the step, from 3.90 in nominal terms to 4.50 in real terms and 5.49 relative to r^*.

Out of sample. Recursive forecasts from 1995 have root mean squared errors of 0.95 (flow) and 0.92 (mixed) times those of the gap alone, with Clark–West statistics of 1.95 and 2.31. The primary family has nine tests, three outcomes in three forms; five reject at the 10 percent level before adjustment, and one survives the Holm correction (the mixed form for unemployment, adjusted $p = 0.09$). The out-of-sample gains are modest.

Payrolls and inflation. For payrolls the in-sample gain is 0.01 in the flow form and the out-of-sample statistic is 2.00; on 1996–2026 the payroll flow is not distinguishable from zero (Section 8.1). For core inflation the index adds nothing (gain 0.00). The claim is about unemployment.

Which periods. In 2009–15, when the policy rate was at its lower bound, the index adds in sample ($t = 1.94$, 84 months) and not significantly out of sample (Clark–West 0.47 in the flow form and 1.30 in the mixed form). In 2022–26 the sample is about three independent years and the coefficient on the index has the opposite sign ($t = -3.31$): the tightening of 2022–23 was not followed by higher unemployment.

5.3 Episodes and Today

Table 6 gives the wedge in three episodes and for September 2026.

Table 5: Does the Index Add to the Funds-Rate Gap?

	Gain of index	t of index	Gain of gap	RMSE ratio	Clark–West
Unemployment gap: Market index, level	0.00	0.4	0.02	1.08	−0.7
Unemployment gap: Market index, flow	0.11	5.5	0.00	0.95	2.0
Unemployment gap: Market index, flow, with the gap in levels	0.12	6.2	0.02	0.92	2.3
Unemployment gap: GS, flow	0.12	5.1	0.00	0.93	2.1
Unemployment gap: FRB/US index, flow	0.10	5.4	0.00	0.99	1.7
Unemployment gap: Market index, flow, 2009–15	0.06	1.9	−0.01	1.08	–
Unemployment gap: Market index, flow, 2022–26	0.03	−3.3	0.00	0.96	–
Payroll growth: Market index, flow	0.01	−2.6	0.01	0.98	2.0
Payroll growth: Market index, flow, with the gap in levels	0.02	−2.5	0.02	0.97	1.6
Core inflation: Market index, flow	0.00	−1.1	0.00	1.03	−0.3

Notes: Local projections of the 12-month change in the unemployment gap, 12-month payroll growth and 12-month core inflation on own lags, the funds-rate gap (effective funds rate less one-year expected inflation less r^*) and the index relative to r^*, 1976–2026 (2020 to March 2021 excluded; HAC standard errors, 18 lags). Gain of index (of gap): increase in adjusted R^2 from adding the index (the gap) to a model that has the other. Level: both in levels; flow: both as 12-month changes; mixed: gap in levels, index as 12-month change. RMSE ratio and Clark–West: recursive out-of-sample forecasts from 1995 of the model with both variables relative to the model with the gap alone (below 1: the index helps; Clark–West above 1.3 is significant at 10 percent); the 2009–15 and 2022–26 rows are in sample.

Table 6: The Index Relative to the Funds-Rate Gap in Three Episodes and Today

	Funds-rate gap	Index	Index less gap [90%]	Kiley r^* [90%]	GS less gap	FRB/US less gap
June 2004 to June 2006	−0.89	−2.18	−1.29 [−1.86, −0.65]	−1.23 [−1.77, −0.62]	−1.21	−0.73
March 2022 to December 2024	0.86	−0.54	−1.40 [−2.22, −0.39]	−1.58 [−2.49, −0.40]	−1.48	−0.04
March 2022 to December 2023	0.09	−0.64	−0.73 [−1.53, 0.29]	−0.90 [−1.77, 0.29]	−0.74	0.62
September 2026	0.46	−0.93	−1.39 [−2.47, −0.22]	−1.55 [−2.66, −0.28]	−1.69	−0.19

Notes: Averages over the window of the funds-rate gap and the market index relative to r^* (Laubach–Williams), each less its 1973–2026 mean; index less gap is in points (the index in two-year-yield points, the gap in funds-rate points). The 90 percent interval takes the 5th and 95th percentiles over 500 bootstrap samples of the announcements (the weights); the Kiley column replaces r^* in the index and in the gap. GS and FRB/US: point estimates only (their weights are held fixed).

In June 2004 to June 2006 the funds-rate gap averaged −0.89 and the index −2.18, a wedge of −1.29 with a 90 percent interval from −1.86 to −0.65; the interval is below zero in 100 percent of the bootstrap samples. With the estimate of Kiley (2020) for r^* the wedge is −1.23. This is the conundrum of Hatzius et al. (2010): conditions stayed easy while the funds rate rose. From March 2022 to December 2024 the wedge is −1.40 (interval −2.22 to −0.39; Kiley −1.58). It depends on the end date: to December 2023 it is −0.73 and the interval includes zero. For September 2026 the funds-rate gap is 0.46 and the index −0.93, a wedge of −1.39 (interval −2.47 to −0.22; negative in 98 percent of the samples). The intervals cover the uncertainty of the weights and the choice of r^* between two estimates. They do not cover the uncertainty of r^* itself or of the funds-rate gap's expected inflation.

5.4 What Carries the Wedge

Equity valuation carries it. Entering the three variables' contributions separately, equity valuation raises the adjusted R^2 for unemployment by 0.13 ($t = 5.2$), the 10-year yield and the rates group by 0.07, and the dollar by 0.00. The index is not a summary of rates. Equity prices also lead the cycle, which Section 6.3 takes up.

6 What the Index Says about Activity

6.1 The Flow, Not the Level

Table 7 enters the level and the 12-month change of each index together. For the market, Goldman Sachs and FRB/US indexes relative to r^* the flow has t-statistics of 5.2, 5.3 and 4.9 for unemployment and the level 0.5, 0.6 and −0.1. The NFCI and ANFCI are the reverse: the level has $t = 3.8$ and 3.0 and the flow 1.0 and 0.6. FCI-G, a flow by construction, carries it in its reading.

The Granger test on the 12 lags of the level in the VAR is strongly significant ($p = 6 \times 10^{-9}$), which looks like a conflict. Table 8 resolves it. Rewriting the 12 lags as the current level and 11 one-month changes, the current level is not significant ($p = 0.17$) and the changes are ($p = 2 \times 10^{-7}$): the test picks up the movement of the level within the year, not where it stands. The result for the level depends on 2008–15. Excluding those years the level has $t = 3.9$ (Table 15).

The lower tail. Following Adrian, Boyarchenko and Giannone (2019), I estimate quantile regressions for payroll growth. In sample the market index's flow shifts the 10th percentile, with a coefficient of −0.63 (90 percent interval −0.96 to −0.08), the same as in the mean. The NFCI level shifts it by −1.14 (−2.22 to −0.56). Out of sample the tail cannot be tested: the evaluation sample has 2 recessions.

6.2 The NFCI

The NFCI and ANFCI beat the common-basis indexes for unemployment at long horizons and for recessions. They are a different index: a level index of credit, risk and leverage, mostly spreads. The

Table 7: Level or Flow? Unemployment and Payrolls a Year Ahead

	t level	t flow	Gain level	Gain flow	t level	t flow
Market index, nominal	0.9	3.9	0.02	0.12	1.2	−2.9
Market index, real	0.6	4.3	0.02	0.13	1.7	−2.9
Market index, relative to r^*	0.5	5.2	0.02	0.14	2.2	−2.9
GS, relative to r^*	0.6	5.3	0.03	0.15	2.1	−3.2
FRB/US index, relative to r^*	−0.1	4.9	0.01	0.13	1.9	−2.9
FCI-G, relative to r^*	2.8	0.5	0.19	0.15	−1.0	−0.7
NFCI	3.8	1.0	0.19	0.12	−1.2	−2.1
ANFCI	3.0	0.6	0.18	0.10	−1.0	−1.5

Notes: Local projections of the 12-month change in the unemployment gap (first four columns) and of 12-month payroll growth (last two) on own lags, the level of the index and its 12-month change entered together, 1976–2026 (HAC standard errors, 18 lags); gain: increase in adjusted R^2 of the index alone over own lags. FCI-G is a flow by construction: its reading is entered as the level and its change as the flow.

Table 8: What the Granger Test on the Level Picks Up

	All 12 lags	Current level (sum of lags)	11 changes
Market index, nominal	1.9e−05	0.27	4.6e−05
Market index, relative to r^*	6.0e−09	0.17	1.5e−07
GS, relative to r^*	3.0e−11	0.15	3.7e−09
FRB/US index, relative to r^*	4.2e−08	0.35	4.6e−07
FCI-G, relative to r^*	6.3e−11	0.00	0.121
NFCI	2.1e−13	0.00	1.6e−04
ANFCI	1.4e−14	0.00	6.6e−04

Notes: p-values of tests in the unemployment equation of the two-variable VAR (12 lags, 1973–2026): the 12 lags of the index jointly (the Granger test); their sum, which is the coefficient on the current level once the lags are rewritten as the level and 11 one-month changes; and the 11 changes jointly.

Table 9: The Market Index and the NFCI Together

	Market flow	t	NFCI level	t
Unemployment gap, 6 months	0.20	4.3	0.16	3.1
Unemployment gap, 12 months	0.22	3.1	0.35	3.6
Unemployment gap, 24 months	−0.04	−0.3	0.67	5.3
Payroll growth, 12 months	−0.23	−1.3	−0.27	−1.3
Recession within 12 months (logit)	0.46	1.3	1.77	4.2

Notes: Coefficients on the 12-month change of the market index relative to r^* and on the level of the NFCI, entered together with own lags (unemployment gap in points per standard deviation of the regressor; HAC standard errors, 12 lags; logit coefficients in the last row).

Table 10: Does the Flow Survive Removing Activity?

Relative to r^*	Unpurged	Purged, full sample	Purged, recursive
Market index	0.57 [0.22, 0.84]	0.32 [0.10, 0.51]	0.19 [0.11, 0.33]
GS	0.71 [0.31, 0.96]	0.43 [0.17, 0.63]	0.25 [0.15, 0.43]

Notes: $\Phi(12)$ with 90 percent block-bootstrap interval from the two-variable VAR (12 lags, January 1983 to September 2026). Purged: each input variable (10-year yield, equity valuation, dollar; GS inputs for GS) is regressed on current and 12 lags of the unemployment gap and 12-month core PCE inflation, on the full sample or recursively (expanding window), and the index is rebuilt from the residuals with unchanged weights.

three variables of the market index explain 0.36 of the NFCI's level, and 0.51 when the Baa–10-year spread is added. Table 9 enters the market flow and the NFCI level together. For unemployment at 6 and 12 months both keep their coefficients ($t = 4.3$ and 3.1 for the market flow, 3.1 and 3.6 for the NFCI). At 24 months only the NFCI remains (5.3, against −0.3), and for recessions the NFCI has $t = 4.2$ against 1.3. The two are complements: the market index is a policy-channel flow in policy units that carries the next six to twelve months, and the NFCI carries the longer horizons and recession risk. It is not a stress or recession monitor.

6.3 Is the Flow Only Activity Reflected Back?

Asset prices respond to expected activity, and Hatzius et al. (2010) purge each indicator of current and lagged growth and inflation so that the index reflects exogenous conditions; the ANFCI does the same. I regress each input on the unemployment gap and core inflation (current and 12 lags), on the full sample and recursively, and rebuild the index from the residuals with the same weights. Table 10 gives the VAR response. The 12-month response of the market index falls from 0.57 to 0.32 with the full-sample purge and 0.19 with the recursive purge, and every interval stays above zero; the Goldman Sachs index is similar. Purging is not identification, and it removes too much when asset prices rightly respond to expected activity. The flow carries both a channel and an anticipation component, and for payrolls it does not survive the purge.

Table 11: The Level of the Index at a Zero Unemployment Gap, 2009–19 Against 1990–2007

	Change in level [90%]	In standard deviations of the index
Market index, nominal	−2.13 [−2.60, −1.57]	−0.52
Market index, real	−1.44 [−1.80, −0.95]	−0.45
Market index, r^* (Laubach–Williams)	0.88 [0.61, 1.42]	0.31
Market index, r^* (Kiley)	0.03 [−0.39, 0.33]	0.01
GS, r^*	1.01 [0.70, 1.52]	0.39
FRB/US index, r^*	0.66 [0.29, 1.05]	0.29

Notes: Intercept of a regression of the index level on the unemployment gap, by period, 2009–19 minus 1990–2007, with a 90 percent block-bootstrap interval (24-month blocks). A level that means the same thing across regimes would show a change near zero.

6.4 The Neutral Rate and the Weights

The neutral rate. The r^* step makes the level of the index mean the same thing across regimes to a degree. At a zero unemployment gap the index level in 2009–19 is 2.13 lower than in 1990–2007 in nominal terms and 1.44 lower in real terms, and 0.88 higher relative to r^* (Laubach–Williams) and 0.03 higher with the estimate of Kiley (2020) (Table 11). The step reverses and shrinks the post-2008 drift but does not make the unemployment response constant across periods (Section 8.1), and the unit restriction it imposes on r^* cannot be tested at this power. For outcomes, the r^* versions have lower out-of-sample error than the nominal versions in 69 of 91 comparisons across seven index families, five outcomes and horizons; 18 Diebold–Mariano tests favor r^* and 2 the nominal version at the 10 percent level, none survives the Holm correction at 5 percent, and the mean error ratio is 0.98. The tests share data, so the counts are descriptive. For unemployment the in-sample gain of the r^* version over the nominal version is 0.030 on 1996–2026 (interval 0.001 to 0.046) and −0.006 before 1996. The outcomes lean toward the r^* basis without proving it; it rests on the argument of Dudley (2010) and Caballero, Caravello and Simsek (2026) that the rate relative to neutral is what matters.

The weights. No alternative weights beat the market weights. Replacing them with the FRB/US weights, the first principal component, equal weights or weights fitted to activity gives mean squared errors 1.04 to 1.07 times those of the market weights for the 12-month change, 25 of 120 paired tests reject at 10 percent before adjustment and 0.00 after. Weights fitted to activity are too imprecise to rank any set. Outcomes cannot settle the weights question; the policy-news weights are not beaten.

7 Do the Indexes Agree?

Table 12 compares the market index with the others at each step.

Table 12: The Market Index and the Other Indexes at Each Step

	Nominal	Real	Relative to r^*
Correlation of levels			
Market and GS	0.76	0.99	0.99
Market and FRB/US	0.98	0.95	0.94
Market and FRB/US with house prices	0.98	0.95	0.94
Market and ANFCI (from 1990)	0.17	0.22	0.22
Market and GS, levels since 1990	0.84	0.97	0.97
Market and FRB/US, levels since 1990	0.85	0.78	0.79
Correlation of 12-month changes			
Market and GS	0.95	0.96	0.98
Market and FRB/US	0.95	0.95	0.96
Correlation of flows			
Forward impetus and FCI-G	0.80	0.85	0.84
Forward impetus and FRB/US growth impetus	0.89	0.88	0.84
FCI-G and FRB/US growth impetus	0.91	0.96	0.98
Average range of the four level indexes (pp)			
1974–2026	2.82	1.16	1.08
1990–2026, zero at the 1990–2026 mean	1.15	0.97	0.88
Latest reading, September 2026			
Market index	−3.34	−2.80	−0.93
GS	−1.36	−2.89	−1.23
FRB/US index	−1.55	−1.04	+0.27
Forward impetus	+0.27	+0.12	+0.18
FCI-G (1-year)	−0.31	−0.11	+0.03

Notes: Monthly. Level indexes (market, GS, FRB/US, FRB/US with house prices) in two-year-yield points (market, GS) or funds-rate points (FRB/US), tighter above zero, relative to the 1973–2026 mean of each variable. FCI-G and the growth impetus of FRB/US in percentage points of GDP growth; forward impetus in index units. Range: highest minus lowest of the four level indexes in a month, averaged. ANFCI converted to two-year-yield points by its response to policy news. Real and r^* versions of GS and FCI-G are my replications with real rates or gaps to r^*.

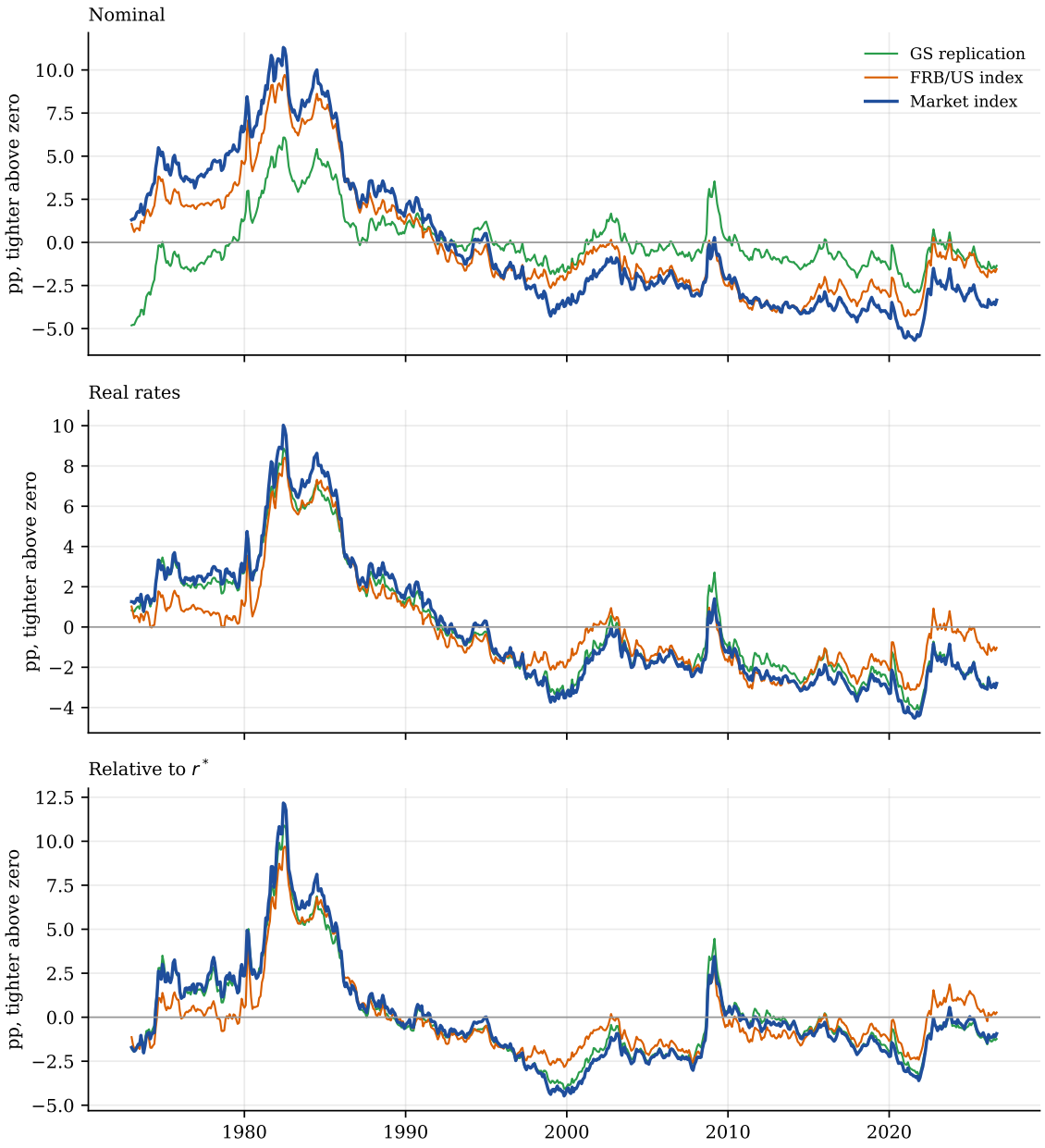


Figure 6: The Level Indexes at Each Step of the Construction

Notes: Monthly, January 1973 to September 2026, relative to the 1973–2026 mean of each variable, tighter above zero. Market index and Goldman Sachs index in two-year-yield points (a policy move that raises the two-year yield 1 percentage point raises each by one point in the announcement window); FRB/US index in funds-rate points. The Goldman Sachs index is my replication with its published weights.

7.1 Levels

The market index and the Goldman Sachs index agree once both are real. With real rates or gaps to r^* the two correlate 0.99 in levels and 0.98 in 12-month changes. In nominal terms the level correlation is 0.76, against 0.98 with the FRB/US index, and the Goldman Sachs index is the one that uses the nominal dollar at that step.

The market index and the FRB/US index agree at every step, a little less at each. The correlation over 1974–2026 is 0.98 in nominal terms, 0.95 in real terms and 0.94 relative to r^*. The 1970s and early 1980s, when all indexes swung widely, dominate these figures; since 1990 the correlations are 0.85, 0.78 and 0.79, and with the Goldman Sachs index 0.97 at the r^* step. Correlations of persistent levels overstate agreement, and the 12-month changes (0.98 and 0.96) are the better guide to whether the indexes move together. Adding house prices to the FRB/US index does not change this (0.94). Both indexes use the real dollar at every step, so the common basis moves the Goldman Sachs index and leaves them where they were; the level correlation with the FRB/US index falls slightly with each step.

The ANFCI is a different object. Its correlation with the market index is 0.22 from 1990. The ANFCI measures stress, mostly spreads and volatility, rather than the price of credit and risk.

The distance between the indexes falls with each step, and stays near a point since 1990. The average range of the four level indexes, each in its own units (two-year-yield points or funds-rate points), is 2.82 points over 1974–2026 in nominal terms, 1.16 in real terms and 1.08 relative to r^*. Over 1990–2026 it is 1.15, 0.97 and 0.88. In 37 percent of the months since 1990 the indexes differ by more than one point. Correlations measure whether the indexes move together. The range measures whether they say the same thing about the level of conditions, and it is the more demanding test. It depends on the units: the market index's standard deviation is 1.28 times that of the FRB/US index, so part of the range is a difference of scale, and the Goldman Sachs conversion is itself estimated (Appendix A). All indexes share the same base period, so the range measures dispersion around each index's own average and not a difference in average level.

7.2 Flows

The market index's forward impetus correlates 0.80 with FCI-G at the nominal step, 0.85 at the real step and 0.84 at the r^* step. Against the published FCI-G the correlation is 0.77 over 1990–2026, and my replication of FCI-G tracks the published series at 0.98. The flow built from FRB/US multipliers agrees with FCI-G more closely (0.98 at the r^* step). The market index's flow comes from a small VAR, not from a structural model, and it agrees with both: its correlation with the FRB/US growth impetus is 0.84.

7.3 Where the Indexes Differ

Table 13 lists the widest and the closest six-month windows at the r^* step, with the gap between the market index and the FRB/US index split by group of variables.

Table 13: Widest and Closest Six-Month Windows, Relative to r^*

Window	Range	Market	GS	FRB/US	FRB/US+HP	Equities	Dollar	Rates	House
Widest									
1977-12 to 1978-05	2.56	2.75	2.28	0.49	0.19	+1.59	+0.33	+0.34	+0.30
1978-11 to 1979-04	2.74	2.26	1.91	0.00	−0.47	+1.58	+0.42	+0.27	+0.47
1979-10 to 1980-03	2.52	3.05	2.84	1.05	0.53	+1.74	+0.29	−0.03	+0.51
Widest since 1990									
1998-10 to 1999-03	1.80	−4.13	−3.54	−2.51	−2.32	−1.37	−0.07	−0.17	−0.19
1999-04 to 1999-09	1.76	−4.10	−3.70	−2.48	−2.35	−1.49	−0.11	−0.03	−0.13
1999-11 to 2000-04	1.75	−4.25	−3.93	−2.57	−2.50	−1.57	−0.11	−0.01	−0.07
Closest									
1986-11 to 1987-04	0.25	0.95	0.94	1.14	1.15	+0.09	−0.16	−0.13	0.00
1989-04 to 1989-09	0.31	−0.07	−0.28	0.00	−0.09	−0.22	+0.11	+0.04	+0.09
1989-10 to 1990-03	0.21	−0.42	−0.57	−0.37	−0.41	−0.27	+0.16	+0.06	+0.05

Notes: Non-overlapping six-month windows ranked by the average range of the four level indexes (averages over the window; relative to the 1973–2026 mean). Gap: market index minus FRB/US index with house prices, by group of variables. The first group ranks all months, the second months from 1990.

Equity valuation accounts for most of the gaps between the market index and the FRB/US index. The widest windows overall fall in 1977–1980, when the market index was about two points tighter than the FRB/US index, with the equity group accounting for about 1.6 points. Equity valuations were then low (CAPE averaged 9.39 over 1977–80 against 22.25 over 1973–2026), and the market index gives each point of valuation more weight (Table 3). Since 1990 the widest windows are in 1998–2001, at the height of the equity boom, when valuations were far above average and the market index was about 1.6 points looser than the FRB/US index, again almost entirely because of equities. In the closest windows, in the late 1980s, the equity and dollar gaps are small. A gap opens when equity valuation or the dollar is far from its average, because the indexes then disagree about how much a given deviation matters.

Today the market index reads −0.93 and the Goldman Sachs index −1.23, while the FRB/US index reads 0.27 (0.16 with house prices), all relative to the 1973–2026 average. The group contributions against the FRB/US index with house prices show why (Figure 7 and Appendix Table A.2). High equity valuations lower the market index by 2.19 points and the FRB/US index by 1.35, and a strong dollar raises them by 0.82 and 1.26. The market index is 0.84 points lower from equities and 0.44 lower from the dollar, out of a total gap of 1.09.

7.4 How Precisely Is the Market Index Estimated?

The market index's weights come from regressions on 253 announcements, and they are estimated with error. I resample the announcements with replacement 500 times, recompute the policy-news factor and the weights each time, and recompute the index and its comparisons (Table 14).

The weights are imprecise. The weight on the 10-year yield has a 90 percent interval from 0.13 to 1.13 around 0.61. The interval for equity valuation runs from −0.07 to −0.03, and for the dollar from 0.02 to 0.11. The FRB/US weights on all three lie inside these intervals, the equity weight

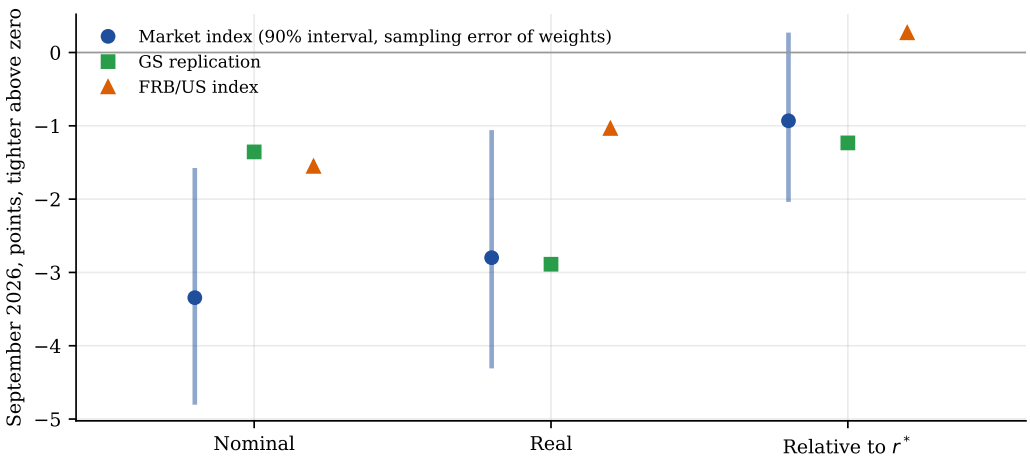


Figure 7: The Reading for September 2026

Notes: Level of each index in September 2026 relative to the 1973–2026 mean, at each step. The bar shows the 5th to 95th percentile of the market index over bootstrap samples of the announcements (Section 7.4).

Table 14: How the Sampling Error of the Weights Affects the Market Index

	Corr. with GS	Corr. with FRB/US	Range, 1990–2026	Latest, market	Latest, GS	Latest, FRB/US
Nominal	0.76 [0.73, 0.78]	0.98 [0.94, 0.99]	1.15 [0.89, 1.87]	−3.34 [−4.77, −1.61]	−1.36	−1.55
Real	0.99 [0.93, 1.00]	0.95 [0.89, 0.99]	0.97 [0.83, 1.48]	−2.80 [−4.28, −1.09]	−2.89	−1.04
Relative to r^*	0.99 [0.91, 0.99]	0.94 [0.88, 0.98]	0.88 [0.76, 1.36]	−0.93 [−2.01, 0.24]	−1.23	+0.27

Notes: Point estimate and 5th to 95th percentile over 500 bootstrap samples of the announcements, each re-estimating the policy-news factor and the weights. Correlations of monthly levels, 1974–2026. Range: average range of the four level indexes since 1990, relative to the 1990–2026 mean. GS and FRB/US are held fixed (their weights are not estimated here).

near the edge: it is lower in absolute value than 7 percent of the bootstrap draws, about 14 percent in a two-sided test. The data do not reject that the model weights equal the market weights. This is a statement about power, not a finding that they are equal, and it treats the FRB/US weights and the VAR as fixed.

The index is better determined than its weights, and its correlations with the others are stable. The correlation with the Goldman Sachs index is 0.99, with a 5th to 95th percentile range from 0.91 to 0.99, and with the FRB/US index 0.94, from 0.88 to 0.98. The level of the reading is not stable. The September 2026 reading relative to r^* is -0.93, with a 90 percent interval from -2.01 to 0.24. The Goldman Sachs reading (-1.23) lies inside that interval. The FRB/US reading (0.27) lies just above it, and the 90 percent interval for the market index minus the FRB/US index runs from -2.28 to -0.03. The sampling error of the weights is therefore of the same size as the differences between the readings, and the Goldman Sachs and FRB/US readings lie on opposite sides of the 1973–2026 average.

8 Robustness and Limits

The results rest on one sample, one set of weights, one estimate of r^* and data revised after the fact. I vary each in turn. Table 15 gives the main checks.

8.1 The Sample: 1996–2026

The weights come from announcements since 1996, so the sample of 1996–2026 is the one on which the index is estimated, and the sample of Note B. I re-estimate the VAR and the regressions from 1996. The response of unemployment after 12 months is 0.65 (0.39 to 0.83) on 1973–2026, and 0.73 (0.26 to 1.00) with 12 lags and 0.58 (0.16 to 0.91) with four lags on 1996–2026: it does not change sign, and the interval is about 1.6 times as wide. The estimate from VARs on shorter spans is 0.64 for 1973–89, 0.39 for 1990–2007, 0.61 for 1996–2007, 0.83 for 1996–2019 and 0.55 for 2009–26. The flow has $t = 4.2$ for unemployment (5.2 on the long sample), with a gain of 0.05 in adjusted R^2 against 0.14; the index adds 0.07 to the funds-rate gap, with a Clark–West statistic of 2.0 (one recession in the out-of-sample period). The payroll flow has $t = -0.8$ and the payroll result does not hold. The r^* step helps on 1996–2026 and not before.

8.2 Weights That Move

The weights define the unit, so I hold them fixed and show how they would drift. Re-estimating on an expanding window of announcements from 2006 keeps the 10-year and dollar weights inside the 90 percent band of the full-sample estimate at every date after 2010; the equity weight is outside at 10 dates in 2011–14, by at most -0.073 against a band edge of -0.070 (Figure 8). A 20-year rolling window fails after 2021, because it drops the announcements of 1996–2005 (the 10-year weight is outside the band at 21 dates). The 10-year weight is imprecise throughout (interval 0.13 to 1.13). The index built with the expanding-window weights correlates 0.97 with the baseline in levels, 0.98

Table 15: The Foundation on 1973–2026 and on 1996–2026

Relative to r^*	$\Phi(12)$ [90%]	t flow	t level	Gain flow	Gain of index given gap	Clark–West	t payroll flow
(a) 1973–2026, 12 lags (main)	0.65 [0.39, 0.83]	5.2	0.5	0.14	0.11	2.0	−2.9
(b) 1996–2026, 12 lags	0.73 [0.26, 1.00]	4.2	−1.2	0.05	0.07	2.0	−0.8
(b) 1996–2026, 4 lags	0.58 [0.16, 0.91]	4.2	−1.2	0.05	0.07	2.0	−0.8
(c) excluding 2008–15	0.56 [0.30, 0.72]	4.5	3.9	0.21	0.13	2.7	−3.4
(d) weights through 2019	0.62 [0.38, 0.78]	5.3	0.7	0.14	0.11	2.1	−3.1
(e) origins from 1974	0.65 [0.39, 0.83]	4.9	0.6	0.12	0.09	2.4	−3.0

Notes: Market index relative to r^*. $\Phi(12)$: VAR response of the unemployment gap after 12 months (2020 dummied; bootstrap interval). t and gain: local projections of the 12-month change of the unemployment gap on the index level and 12-month change together (HAC, 18 lags); gain is the gain in adjusted R^2 of the flow over own lags. Gain of index given gap and Clark–West: as Table 5, flow form. Row (b) uses only origins from 1996 (out-of-sample evaluation from December 2004, one recession); (c) drops 2008–15 (dummies in the VAR, origins touching the window in the regressions); (d) weights from announcements through 2019; (e) regression origins from 1974. Payroll flow: unemployment is the claim, payrolls are not.

in 12-month changes and 0.98 in forward impetus, and the VAR response is 0.52 against 0.65. The weights drift less than their own sampling error.

8.3 Data as Known at the Time

Nothing in the baseline is real time. I rebuild the unemployment and payroll inputs from ALFRED vintages (unemployment and payrolls from 1977, the CBO natural rate from 2011) and the r^* input from the New York Fed's real-time Laubach–Williams estimates (from 2005). The weights, the Kiley r^*, the earnings in CAPE and expected inflation are not real time, so this is pseudo real time in the macro variables and in r^* from 2005, and not in the weights. Table 16 gives the unemployment results. Vintage data leave them unchanged ($t = 5.5$ against 5.5, Clark–West 1.97 against 1.96); with the vintage natural rate from 2011 the flow has $t = 3.1$; and with real-time r^* from 2006 the gain falls from 0.065 to 0.019 ($t = 2.2$), with the mixed form no longer significant ($t = 1.0$). Real-time r^* weakens the flow.

8.4 The Neutral Rate, the Dynamics, the Weights and the Zero Point

Table 17 varies the construction. In every row the market index's correlation is at least 0.91 with the Goldman Sachs index and at least 0.91 with the FRB/US index, and the 90 percent interval of the yardstick lies above zero.

With the estimate of Kiley (2020) the market index reads −1.40 in September 2026 against −0.93, and the Goldman Sachs and FRB/US readings move by less than half a point. With four lags the yardstick is 0.40. Re-estimating the weights with the two-year yield as the policy news, with announcements through 2019, or without the lower-bound years (Appendix Table A.3) moves the September 2026 reading from −1.49 to 0.41, a spread of 1.91 points, as wide as the bootstrap interval of Section 7.4. The readings follow the equity weight, which is −0.027, −0.061 and −0.039 against −0.051 in the baseline. With 1990–2026 as the base period the market index reads 0.45, the Goldman Sachs index −0.001 and the FRB/US index 1.19, shifts of about a point.

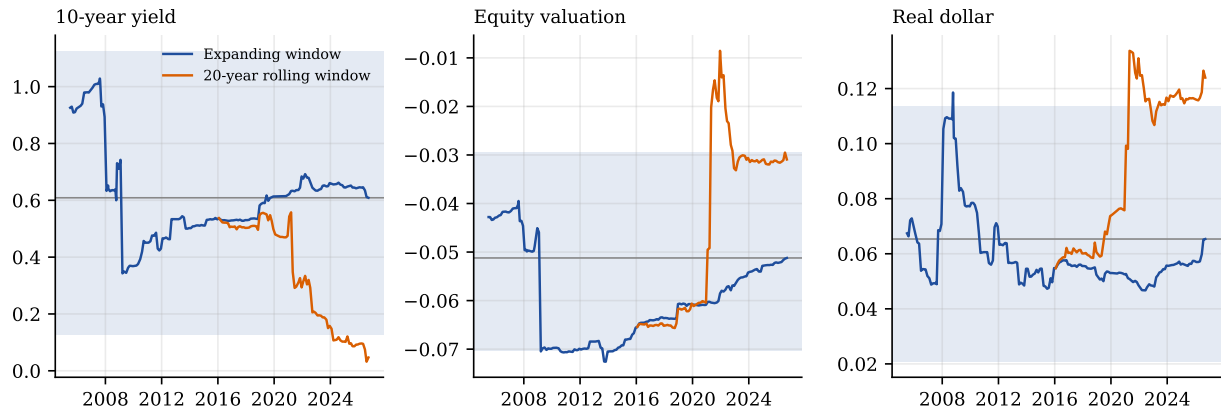


Figure 8: The Weights on an Expanding and a Rolling Window

Notes: Weights of the market index re-estimated on announcements to each date (expanding window, from the 80th announcement) and on the latest 20 years (rolling). Shaded: 90 percent bootstrap interval of the full-sample weights; the horizontal line is the full-sample weight.

Table 16: The Unemployment Results with Vintage Data and Real-Time r^*

Unemployment gap	n	t flow	Gain flow	Clark–West	t mixed	Gain mixed
1976-2025: Revised data	548	5.5	0.11	2.0	6.2	0.12
1976-2025: Vintage data	548	5.5	0.11	2.0	6.4	0.12
2011-2025: Revised data	137	2.6	0.02	–	2.5	0.02
2011-2025: Vintage data	137	2.6	0.02	–	2.5	0.02
2011-2025: Vintage data and NROU	137	3.1	0.04	–	3.1	0.04
2006-2025: Revised data, full-sample r*	166	3.2	0.06	–	2.2	0.02
2006-2025: Revised data, real-time r*	166	2.2	0.02	–	1.0	0.00
2006-2025: Vintage data, real-time r*	166	2.4	0.02	–	1.2	0.00

Notes: Market index relative to r^*, index given funds-rate gap (flow and mixed forms of Table 5), same designs on revised data and on data as known at the time: vintage unemployment rates (ALFRED) with the vintage CBO natural rate (from 2011) and the New York Fed's real-time Laubach–Williams r^* (from 2005; published estimates have a gap from August 2020 to May 2023). The weights, the Kiley r^*, the earnings in CAPE and expected inflation are not real time. Clark–West only for the long sample.

Table 17: Variations of the Construction

	Market–GS	Market–FRB/US	Range	Latest	Φ(12) [90%]	Flow–FCI-G [5–95%]
Baseline: Laubach-Williams r*, 12 lags, VAR 1973-2026	0.99	0.94	0.88	−0.93	0.65 [0.39, 0.83]	0.84 [0.52, 0.90]
Kiley r*	0.99	0.95	0.79	−1.40	0.61 [0.33, 0.80]	0.85 [0.56, 0.91]
4 VAR lags	0.99	0.94	0.88	−0.93	0.40 [0.21, 0.58]	0.72 [0.48, 0.90]
VAR estimated from 1990	0.99	0.94	0.88	−0.93	0.62 [0.21, 0.92]	0.77 [−0.03, 0.88]
VAR estimated to 2019	0.99	0.94	0.88	−0.93	0.68 [0.42, 0.85]	0.84 [0.54, 0.91]
Weights: policy news is the 2-year yield	0.91	0.96	0.96	+0.41	0.52 [0.23, 0.71]	0.76 [0.31, 0.85]
Weights: announcements through 2019	0.99	0.91	1.06	−1.49	0.62 [0.38, 0.78]	0.84 [0.55, 0.90]
Weights: no announcements at the lower bound	0.95	0.95	1.05	−0.22	0.52 [0.27, 0.69]	0.79 [0.43, 0.87]

Notes: Baseline: relative to r^* (Laubach–Williams), 12 lags, VAR from 1973, announcement-window weights from the Nakamura–Steinsson factor. Market–GS and Market–FRB/US: correlations of monthly levels, 1974–2026. Range: average range of the four level indexes, 1990–2026. Latest: market index, September 2026. Φ(12): unemployment response after 12 months with 90 percent block-bootstrap interval. Flow–FCI-G: correlation of the forward impetus with FCI-G (1-year), with the 5th to 95th percentile over bootstrap draws of the response. In the three weights variations GS and FRB/US are the baseline ones.

8.5 Limits of the Design

The r^* step is the largest effect in the paper, and the outcomes lean toward it without proving it. The uncertainty of r^* is not in the intervals, and r^* is not neutral to outcomes, because the Laubach–Williams model uses output, inflation and unemployment. The weights come from announcements since 1996 and are applied from 1973; they assume that the responses reflect policy rather than information effects (Nakamura and Steinsson, 2018), and if policy came to target financial conditions the responses could change, the Lucas-critique concern that Hatzius et al. (2010) raise for all indexes. The bootstrap treats the FRB/US and Goldman Sachs weights, the Goldman Sachs conversion and the VAR as fixed. The sample has five recessions, two out of sample, and 12-month outcomes overlap, so the counts of tests are descriptive. The VAR is linear and describes the mean. The Goldman Sachs index and FCI-G are my replications, and the FRB/US index is my construction at one horizon. The readings for September 2026 use r^* through 2026Q2 and interpolate October 2025.

9 Discussion

Three points follow for reading the stance of policy.

Financial conditions add to the policy rate, through changes. The funds-rate gap tells little about unemployment a year ahead once the index is in, and the index tells much once the gap is in. What it tells is in how conditions have moved over the year, in policy units and relative to r^*, not in where they stand. In 2004–06 and in 2022–24 the index read more than a point looser than the gap, and in September 2026 the wedge is −1.39. These are readings of the stance with intervals, and the stance from the policy rate alone misses them.

The index is a channel measure, not a forecaster or a stress monitor. It carries the next six to twelve months of unemployment, in a form that survives purging of lagged activity at a third to a half of its size. It does not carry inflation, payrolls on 1996–2026 or recession risk, and the NFCI carries the longer horizons. Out-of-sample gains are modest.

The neutral rate changes the readings. In nominal terms the market index reads −3.34 points for September 2026, relative to r^* −0.93, and the step reverses the picture after 2008. The indexes agree on direction (12-month changes correlate 0.98 with the Goldman Sachs index and 0.96 with the FRB/US index) and differ in level by about 1.08 point on average, of the order of the sampling error of the market index's weights. Outcomes lean toward the r^* basis, and the policy-news weights are not beaten.

What the results do not show. There is no ground truth for financial conditions. Zero is the 1973–2026 average of each variable, not a neutral level of conditions. The index measures the policy-equivalent tightening and treats movements that policy did not cause as if it had; the unemployment response is the claim, and the index carries both a channel and an anticipation component.

10 Conclusion

Financial conditions add to the stance of policy beyond the policy rate. Measured in policy units and relative to the neutral rate, the market index explains unemployment a year ahead beyond the real funds rate relative to r^*, through its 12-month change, and not inflation. It complements the NFCI, which carries the longer horizons and recession risk, and it holds on 1996–2026 at smaller size. The wedge between the index and the funds-rate gap was about 1.29 points in 2004–06 and 1.40 points in 2022–24. On a common basis the indexes agree on direction and differ by about a point in level, and the neutral rate changes the reading most. The most durable finding is the wedge and what it says for unemployment, and the answer to what a statement about the stance should say is a range, relative to r^*.

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Appendix A Construction and Estimation Details

Data. All series are monthly averages from January 1973 to September 2026, from the public sources in Table A.1. The BBB spread is Moody's Baa yield less the 10-year yield. Months in which the CPI and unemployment releases are missing (October 2025) are interpolated. The unemployment gap is the unemployment rate less the Congressional Budget Office's natural rate, placed in the middle month of each quarter and interpolated. Expected inflation is the Cleveland Fed's estimate at one and ten years from 1982 and the Livingston Survey's 12-month CPI expectation before, shifted to the Cleveland level by the mean difference over 1982–84 and blended over 1982. CAPE is Shiller's series spliced over 1990–92 in logs to the monthly average of a daily series. The real broad dollar is spliced in 2006 in logs, as are the nominal dollar and the S&P 500. Case–Shiller home prices before 1987 are Shiller's series.

Market index. Policy news is the first principal component of MP1, MP2, ED2, ED3 and ED4 in the monetary-event windows, scaled by its regression on the two-year yield change so that one unit moves the two-year yield by one percentage point. Unscheduled events with no change in MP1 and March–April 2020 are dropped. Each of the 10-year yield change, the S&P 500 return and the dollar (DXY) return in the window is regressed on the factor. The residual covariance Σ of the three regressions gives $w = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$. The S&P 500 and DXY weights apply to 100 times the log of CAPE and of the real broad dollar, as percent changes in the window and in the index levels agree to first order.

Goldman Sachs index. The weights are the published ones: policy rate 4.4, 10-year yield 45.1, BBB spread 39.6, equity valuation -4.9 and dollar 6.0 percent, applied to the demeaned inputs. The conversion to the market index's units is the inverse of the response of a daily replication of the index to policy news, measured from the day before to the day after each announcement because its dollar input is a noon series (1.04 points per unit of news, standard error 0.15).

FCI-G. I use the published multipliers of the Board's release. Each of the seven variables enters as the change over three months of its three-month average (rates, in points), 100 times the three-month log change (stock and house prices) or 100 times the log change of the three-month average (the dollar), with the multipliers of the first four quarters for the one-year lookback and the first twelve for the three-year lookback. This replication tracks the published series at 0.98 in monthly levels from 1990. At the r^* step the stock term is the change in the earnings-yield gap.

FRB/US index. The weights are the cumulative GDP responses at four quarters to a persistent unit shift in each variable, simulated in FRB/US with VAR-based expectations, divided by the response to a persistent one percentage point rise in the funds rate. The weight on the policy rate is its direct effect with expectations of future rates held fixed. House prices enter as 100 times the log of the price-to-rent ratio relative to its trailing 15-year average. The growth impetus applies the differences of the quarterly responses at horizons $m/3 + 4$ and $m/3$ to the changes of the last 12 months, the form of FCI-G.

NFCI and ANFCI. The weekly index at the last observation of each month; the conversion is the response of the weekly change to the sum of policy news in the week (Friday-dated weeks;

Table A.1: Data Sources

Series	Frequency	Source
Federal funds rate; 10-year yield; Baa yield; 30-year mortgage rate	Monthly	FRED FEDFUNDS, GS10, BAA, MORTGAGE30US
Real broad dollar; nominal broad dollar	Monthly	FRED RTWEXBGS, TWEXBPA; DTWEXBGS, TWEXBMTH (spliced in 2006)
CPI; CPI rent; Case–Shiller home prices; S&P 500	Monthly	FRED CPIAUCSL, CUUR0000SEHA, CSUSHPISA, SP500; Shiller's data before the FRED start
Unemployment rate; natural rate of unemployment	Monthly; quarterly	FRED UNRATE; CBO, FRED NROU
Expected inflation, 1 and 10 years	Monthly	Cleveland Fed, FRED EXPINF1YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed)
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
r^*, Kiley	Quarterly	One-sided estimates of the author
CAPE	Monthly	Shiller, spliced to the monthly average of a daily CAPE constructed by the repository's daily workflow
NFCI, ANFCI	Weekly	Chicago Fed, FRED
FCI-G and its multipliers	Monthly	Federal Reserve Board, FEDS Notes data release
FRB/US simulations	Quarterly	Run for this paper's companion work with the public FRB/US package
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)

heteroskedasticity- and autocorrelation-robust standard errors with four lags).

Gaps to r^*. For the policy rate, the 10-year yield and the Baa and mortgage yields, the input is the real rate less r^*. Equity valuation becomes $-(100/\overline{E/P})(E/P - r^*)$, where $E/P = 100/\text{CAPE}$ and $\overline{E/P}$ is its 1973–2026 mean, so that a one-point rise of CAPE (in logs) has the same weight as before when E/P is at its mean. r^* is the one-sided, full-sample estimate (New York Fed's Laubach–Williams estimate, data through 2026Q2) placed in the middle month of each quarter.

VAR and bootstrap. The VAR has a constant, the 12 monthly dummies for 2020 and p lags of the unemployment gap and the index. Because unemployment is first, the response of unemployment to an innovation that raises the index by one unit on impact is the moving-average coefficient $\Phi(h)$ and does not depend on the residual covariance. The bootstrap resamples blocks of 12 consecutive residual pairs, simulates the VAR recursively from the actual initial values with the dummies fixed, and re-estimates; bands are percentiles of the 500 draws. The same draws give the intervals of the flow's correlation with FCI-G. The weights bootstrap resamples announcements with replacement and recomputes the policy-news factor and the weights each time. Draws use the seed in `lib/config.py`.

Table A.2: Contributions to the Readings for September 2026, Relative to r^*

	Rates	Equities	Dollar	Other	Total
Market index	+0.44	−2.19	+0.82	0.00	−0.93
GS	+0.06	−2.02	+0.72	0.00	−1.23
FRB/US with house prices	+0.37	−1.35	+1.26	−0.11	+0.16

Notes: Contribution of each group of variables to the index in September 2026 (weight times the variable relative to its 1973–2026 mean), in points; rates include policy rate, 10-year yield and spreads, other is house prices. Equity valuation is the earnings-yield gap to r^*.

Table A.3: Weights Under Other Policy-News Series and Event Samples

	Events	w 10-year	w equities	w dollar	ψ 10-year	ψ equities	ψ dollar
Baseline: Nakamura–Steinsson factor	253	0.609	−0.051	0.065	0.53	−7.78	4.22
Policy news is the 2-year yield change	253	0.925	−0.027	0.070	0.58	−6.09	4.22
Announcements through 2019	201	0.613	−0.061	0.053	0.56	−7.50	3.83
No announcements December 2008–November 2015	197	0.986	−0.039	0.058	0.49	−7.82	3.69

Notes: Weights w and responses ψ as in Table 2, re-estimated on each sample or with the window change of the two-year yield as the policy-news measure. In each case the weights applied to the responses sum to 1.

Table A.4: The VAR in All Cases

Case	Lags	Sample	Granger F	p	$\Phi(6)$	$\Phi(12)$ [90%]	$\Phi(24)$ [90%]	Months	Flow–FCI-G
Nominal	12	1973–2026	3.74	<0.0001	0.29	0.57 [0.29, 0.77]	0.66 [0.32, 0.88]	4–36	0.80
Nominal	4	1973–2026	4.62	0.0010	0.21	0.27 [0.10, 0.45]	0.27 [0.13, 0.51]	3–36	0.64
Real	12	1973–2026	3.67	<0.0001	0.34	0.61 [0.32, 0.82]	0.68 [0.29, 0.95]	3–36	0.85
Real	4	1973–2026	5.54	0.0002	0.26	0.32 [0.13, 0.50]	0.32 [0.15, 0.56]	3–36	0.69
r^*	12	1973–2026	5.30	<0.0001	0.40	0.65 [0.39, 0.83]	0.64 [0.28, 0.89]	1–36	0.84
r^*	4	1973–2026	9.98	<0.0001	0.32	0.40 [0.21, 0.58]	0.38 [0.22, 0.62]	1–36	0.72
r^* (Kiley)	12	1973–2026	4.63	<0.0001	0.37	0.61 [0.33, 0.80]	0.63 [0.27, 0.91]	3–36	0.85
r^* (Kiley)	4	1973–2026	7.76	<0.0001	0.29	0.36 [0.17, 0.54]	0.34 [0.18, 0.58]	1–36	0.70
r^*, from 1990	12	1990–2026	2.91	0.0006	0.37	0.62 [0.21, 0.92]	0.52 [−0.01, 0.88]	3–23	0.77
r^*, to 2019	12	1973–2019	5.42	<0.0001	0.42	0.68 [0.42, 0.85]	0.66 [0.30, 0.92]	1–35	0.84
r^*, from 1990	4	1990–2026	5.63	0.0002	0.38	0.55 [0.18, 0.85]	0.62 [0.21, 0.96]	3–36	0.86
r^*, to 2019	4	1973–2019	9.89	<0.0001	0.34	0.41 [0.21, 0.60]	0.38 [0.21, 0.63]	1–36	0.68
r^*, weights: 2-year news	12	1973–2026	3.55	<0.0001	0.32	0.52 [0.23, 0.71]	0.49 [0.13, 0.79]	3–30	0.76
r^*, weights: to 2019	12	1973–2026	5.64	<0.0001	0.37	0.62 [0.38, 0.78]	0.60 [0.28, 0.82]	1–36	0.84
r^*, weights: no lower bound	12	1973–2026	4.11	<0.0001	0.32	0.52 [0.27, 0.69]	0.50 [0.17, 0.76]	3–33	0.79

Notes: As Table 4. Months: first and last month with the 90 percent band above zero. Flow–FCI-G: correlation of the forward impetus with the FCI-G of the same step, 1974–2026.

Table A.5: Checks: Kiley r^* and 4 Lags, Correlations with the Baseline

	Levels	12-month changes
Market index	0.96	0.96
GS	0.96	0.97
FRB/US index	0.96	0.97
FCI-G, 1-year	0.97	0.98
Market forward impetus	0.96	
Forward impetus, 4 against 12 lags	0.96	

Notes: Correlations of monthly series, 1973–2026.

Table B.1: Unemployment Gap 12 Months Ahead on the Current Index Reading, Relative to r^*

	Coefficient	Gain in adj. R^2	RMSE ratio	1995–2007	2008–19	2021–25	DM t
Market index forward impetus	0.50 (0.09)	0.11	0.89	0.89	0.86	1.14	−1.12
FCI-G, 1-year lookback	0.70 (0.10)	0.14	0.87	0.87	0.85	1.09	−1.05
FRB/US growth impetus	0.30 (0.05)	0.12	0.93	0.94	0.91	1.09	−0.62
Market index (level)	0.07 (0.05)	0.01	1.06	0.94	1.12	0.89	1.00
GS (level)	0.09 (0.06)	0.02	1.05	0.90	1.11	0.86	0.82
FRB/US index (level)	0.07 (0.06)	0.01	1.05	0.90	1.11	0.93	0.97

Notes: Local projection $u_{t+12} = a + b\,u_t + c\,x_t$, 1976–2026; coefficient c (HAC standard error); gain in adjusted R^2 over u_t alone. Out of sample: recursive estimates, forecast origins 1995–2025; ratio of root mean squared errors to the projection on u_t alone (below 1: the index helps), also for three subperiods; DM: Diebold–Mariano t (negative: index better). Months touching 2020 to March 2021 excluded.

Table B.2: Unemployment Gap 12 Months Ahead, Real Versions

	Coefficient	Gain in adj. R^2	RMSE ratio	1995–2007	2008–19	2021–25	DM t
Market index forward impetus	0.50 (0.12)	0.09	0.94	0.90	0.92	1.23	−0.74
FCI-G, 1-year lookback	0.81 (0.14)	0.13	0.91	0.87	0.88	1.17	−0.83
FCI-G, 3-year lookback	0.71 (0.12)	0.13	0.93	0.85	0.94	1.07	−0.73
FRB/US growth impetus	0.32 (0.06)	0.10	0.98	0.99	0.96	1.13	−0.21
Market index (level)	0.05 (0.04)	0.01	1.06	0.91	1.12	0.89	0.87
Market index current pressure	0.02 (0.04)	0.00	1.06	0.97	1.09	0.99	1.37
GS (level)	0.07 (0.05)	0.02	1.05	0.89	1.11	0.85	0.69
FRB/US index (level)	0.06 (0.06)	0.01	1.05	0.89	1.11	0.92	0.84

Notes: As Table B.1.

Appendix B Activity Regressions on Each Index and Step

Tables B.2 and B.3 repeat Table B.1 for the real and nominal versions. The flows carry the information at every step, and in each of the three flows the gain in adjusted R^2 is at least as large for the r^* version as for the real version.

Appendix C Current Pressure

The level of the market index is the eventual effect of current conditions. The part already passed through to unemployment, which I call current pressure, weights past levels by the VAR's response profile and lags the level by several months (Figure C.1). It correlates 0.95 with the level and adds nothing to the year-ahead regressions (Tables B.1 to B.3).

Table B.3: Unemployment Gap 12 Months Ahead, Nominal Versions

	Coefficient	Gain in adj. R^2	RMSE ratio	1995–2007	2008–19	2021–25	DM t
Market index forward impetus	0.39 (0.12)	0.07	0.95	0.91	0.92	1.25	−0.76
FCI-G, 1-year lookback	0.76 (0.16)	0.10	0.90	0.89	0.90	1.00	−1.08
FCI-G, 3-year lookback	0.78 (0.13)	0.12	0.90	0.81	0.93	0.87	−1.19
FRB/US growth impetus	0.30 (0.07)	0.09	0.96	0.94	0.94	1.20	−0.48
Market index (level)	0.04 (0.03)	0.01	1.09	0.96	1.15	0.88	1.03
Market index current pressure	0.03 (0.04)	0.00	1.09	0.97	1.14	0.93	1.13
GS (level)	0.21 (0.08)	0.05	1.02	0.85	1.10	0.67	0.32
FRB/US index (level)	0.05 (0.04)	0.01	1.09	0.90	1.17	0.89	0.98

Notes: As Table B.1.

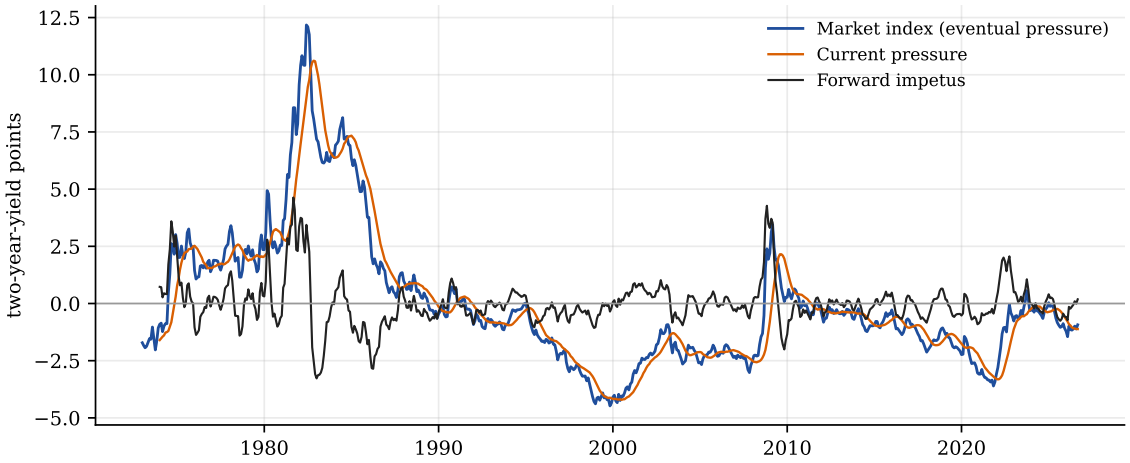


Figure C.1: Market Index: Level, Current Pressure and Forward Impetus

Notes: Relative to r^*, 12-lag VAR, in two-year-yield points.